

# Interspecific competition

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## Learning objectives

1. Know the processes influencing and outcomes of interspecific competition
2. Understand the difference between phenomenological and mechanistic competition models
3. Perform an invasion analysis

## Processes

- interspecific competition: one species has reduced growth rate in presence of another species
- competition *intra* and *inter* enables multiple species to exist in community - diversity maintained
- $\text{intra} > \text{inter}$

*species is limiting itself more than others- this makes sense because if multiple species are going to coexist, we can't have a species that has unrestricted growth, it'll just use up all the resources*

- stabilizing mechanisms:

*these are all stabilizing mechanisms- they make it so multiple species maintained locally*

- niche differentiation - fitness maximized with different resources or levels

*somewhat related is storage effect, more a temporal mechanism that acts on multiple generations*

- storage effect - environmental fluctuations, store seeds in good years, other plants grow in bad years

*this assumes that species have different good years and bad years, ties back to niche differentiation, but more about storing success*

- relative nonlinearity - relationship between fitness and resource level not all same *in other words, not all linear, they have different shapes*

## Modeling interspecific competition

*add to beverton-holt, previously:*

$$N_{i,t+1} = \frac{\lambda N_{i,t}}{1 + \alpha_{ii} N_{i,t}}$$

*add a second species to restrict population growth*

- species i and j
- $\alpha_{ij}$  = effect of j on i

$$N_{i,t+1} = \frac{\lambda N_{i,t}}{1 + \alpha_{ii} N_{i,t} + \alpha_{ij} N_{j,t}}$$

*we just added another term in the denominator, added subscripts to clarify which population we're modeling*

- phenomenological = just effects ( $\alpha$ )
- mechanistic model = models resource levels or processes

*next week we'll talk about some mechanistic models*

## Outcomes

- competitive exclusion - one species outcompetes other species
- coexistence - stabilize and both persist in long term *may have lower population growth or sizes compared to those without competitor, but if they persist, it's coexistence*
- priority effects - depends on species arrival *whichever arrives in the community first*

## Invasion analyses

*I said you have coexistence when both persist in long term*

*but a time series of populations alone can't tell you if there's coexistence*

*plot example with 2 populations fluctuating over time*

*this looks like coexistence, but what if transient dynamics, decline in future?*

- both species increase from low density
- long-term average population growth rate
- $\log \frac{N_{t+1}}{N_t} > 0$  for each species - coexistence,  $< 0$  competitive exclusion